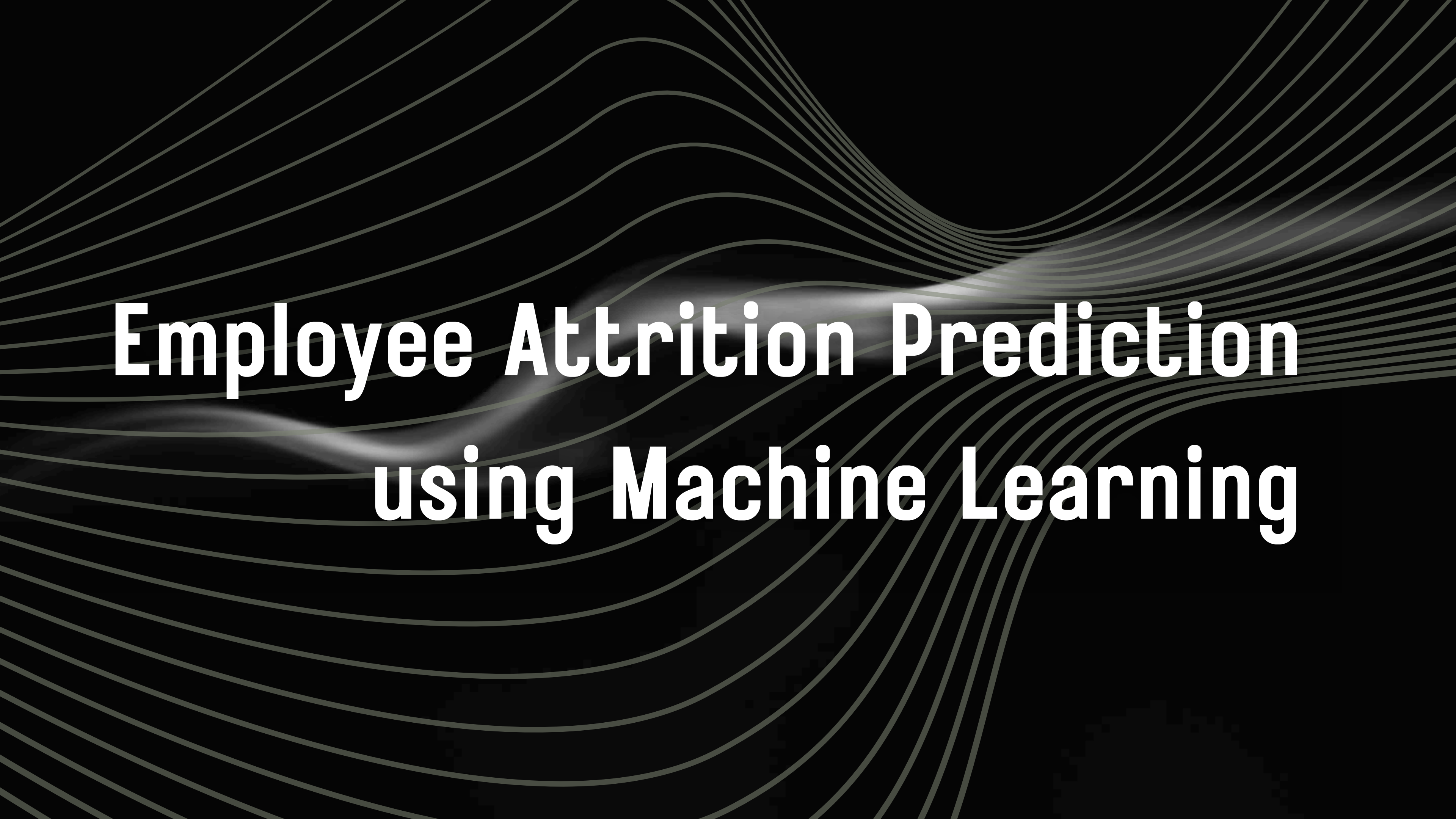




Employee Attrition Prediction using Machine Learning

Problem Statement

Employee attrition is one of the biggest challenges faced by organizations today. Every time an experienced employee leaves, the company spends additional time and money on hiring, onboarding, and training a replacement. Besides financial costs, organizations also lose valuable knowledge, experience, and productivity.

Traditionally, HR departments rely on interviews and manual observations to understand why employees leave. However, these methods are often reactive rather than proactive. This project aims to solve this problem by using machine learning to identify employees who may be at risk of leaving before they actually resign.

Project Objectives

The main objectives of this project are:

- Predict whether an employee will leave the company.
- Analyze the major factors influencing employee attrition.
- Compare different machine learning classification algorithms.
- Identify the most important features responsible for employee turnover.
- Provide practical HR recommendations based on the analysis

Dataset Overview

- IBM HR Analytics Employee Attrition Dataset
- Total Employees: 1,470
- Target Variable: Attrition (Yes/No)

Features include:

- Age
- Department
- Job Role
- Monthly Income
- Job Satisfaction
- Work–Life Balance
- Overtime
- Years at Company

Include:

- Dataset shape
- Number of numerical and categorical features
- Attrition rate

Data Preprocessing

Before building the machine learning models, the dataset was cleaned and prepared.

The preprocessing steps included:

- Loading the dataset using Pandas.
- Checking for missing values.
- Removing unnecessary columns such as EmployeeNumber, Over18, StandardHours, and EmployeeCount because they do not contribute to prediction.
- Converting the target variable Attrition from Yes/No to 1/0.
- Applying One-Hot Encoding to categorical variables such as Department, Job Role, and Business Travel.
- Standardizing numerical features using StandardScaler.
- Splitting the dataset into 80% training data and 20% testing data.

Proper preprocessing improves model performance and ensures consistent results.

Machine Learning Models

Exploratory Data Analysis was performed to understand employee behavior and identify important patterns.

The analysis included:

- Attrition rate across different departments.
- Attrition rate across job roles.
- Monthly income comparison between employees who stayed and those who left.
- Relationship between work–life balance and employee attrition.
- Attrition trend based on years at the company.

The visualizations clearly showed that employee turnover is influenced by several workplace and personal factors rather than salary alone.

Machine Learning Models

Three different machine learning classification algorithms were trained and compared.

Logistic Regression :

This model serves as the baseline classifier. It is simple, fast, and highly interpretable, making it suitable for HR professionals who need understandable predictions.

Random Forest Classifier :

Random Forest combines multiple decision trees to improve prediction accuracy and reduce overfitting. It is capable of handling complex relationships among variables.

Gradient Boosting Classifier :

Gradient Boosting builds decision trees sequentially, where each new tree corrects the errors of the previous one. It generally provides very strong predictive performance.

Using multiple models allowed a fair comparison to identify the most suitable model for employee attrition prediction.

Model Evaluation

The models were evaluated using several performance metrics:

- Accuracy
- Precision
- Recall
- F1–Score
- ROC–AUC Score
- Confusion Matrix

Accuracy measures overall correctness, while Precision indicates how many predicted attrition cases were actually correct. Recall measures how effectively the model identifies employees who truly leave. The F1–Score balances Precision and Recall, and ROC–AUC evaluates the model's ability to distinguish between employees who stay and those who leave.

After comparing all models, the best–performing model was selected based on these evaluation metrics.

Feature Importance

Feature importance analysis was conducted to identify the variables that most strongly influence employee attrition.

The most important features included:

- Overtime
- Monthly Income
- Job Satisfaction
- Age
- Years at Company
- Total Working Years
- Work–Life Balance
- Environment Satisfaction
- Distance From Home
- Stock Option Level

These features provide valuable insights into employee behavior and help HR focus on the most influential factors affecting retention.

Business Insights and Recommendations

Based on the analysis, several important business insights were obtained.

Employees who frequently worked overtime and reported lower job satisfaction were more likely to leave the organization. Lower monthly income also increased the likelihood of attrition, but salary alone was not the only deciding factor. Employees with poor work-life balance and fewer years at the company showed higher turnover.

Recommendations:

1. Reduce excessive overtime by improving workload distribution and workforce planning.
2. Conduct regular employee engagement and satisfaction surveys to identify concerns early.
3. Provide career development programs, promotions, and skill enhancement opportunities.
4. Focus retention efforts on departments and job roles with higher attrition rates.
5. Review compensation and recognition programs for high-performing employees.

Limitations and Future Scope

Although the model performs well, it has certain limitations.

It relies only on historical HR data and cannot capture personal reasons such as family commitments, health issues, external job opportunities, or career aspirations. Therefore, the predictions should be used as decision-support rather than replacing HR judgment.

Future improvements could include:

- Collecting more recent employee data.
- Using advanced machine learning techniques such as XGBoost or LightGBM.
- Applying SMOTE to handle class imbalance.
- Deploying the model as a web application for real-time HR use.
- Integrating employee feedback and survey data for better predictions.

Conclusion

This project successfully demonstrated how machine learning can be applied to predict employee attrition and support strategic HR decision-making.

By comparing multiple classification models, identifying important features, and generating actionable business recommendations, the project provides a practical solution for reducing employee turnover.

Using predictive analytics, organizations can improve employee satisfaction, reduce hiring costs, retain valuable talent, and create a more productive work environment.



THANK YOU